# Static response of a magnetically active fluid (10 points)

## Part A: Static Testing (1.6 points)

### Magnetic interaction: force on a ferrofluid

**A.1** (0.8 pt)
Diagram:

$z =$

**A.2** (0.8 pt)

Density difference $\Delta\rho =$

## Part B: Magnetic interaction: surface tension of a ferrofluid (1.2 points)

**B.1** (0.6 pt)
$z_{\text{crit}} =$

$\lambda =$

**B.2** (0.6 pt)
$\sigma =$

## Part C: Optical surface characterisation: non-spiking regime (4.1 points)

### Optical surface characterisation: non-spiking regime

**C.1** (0.6 pt)
Diagram

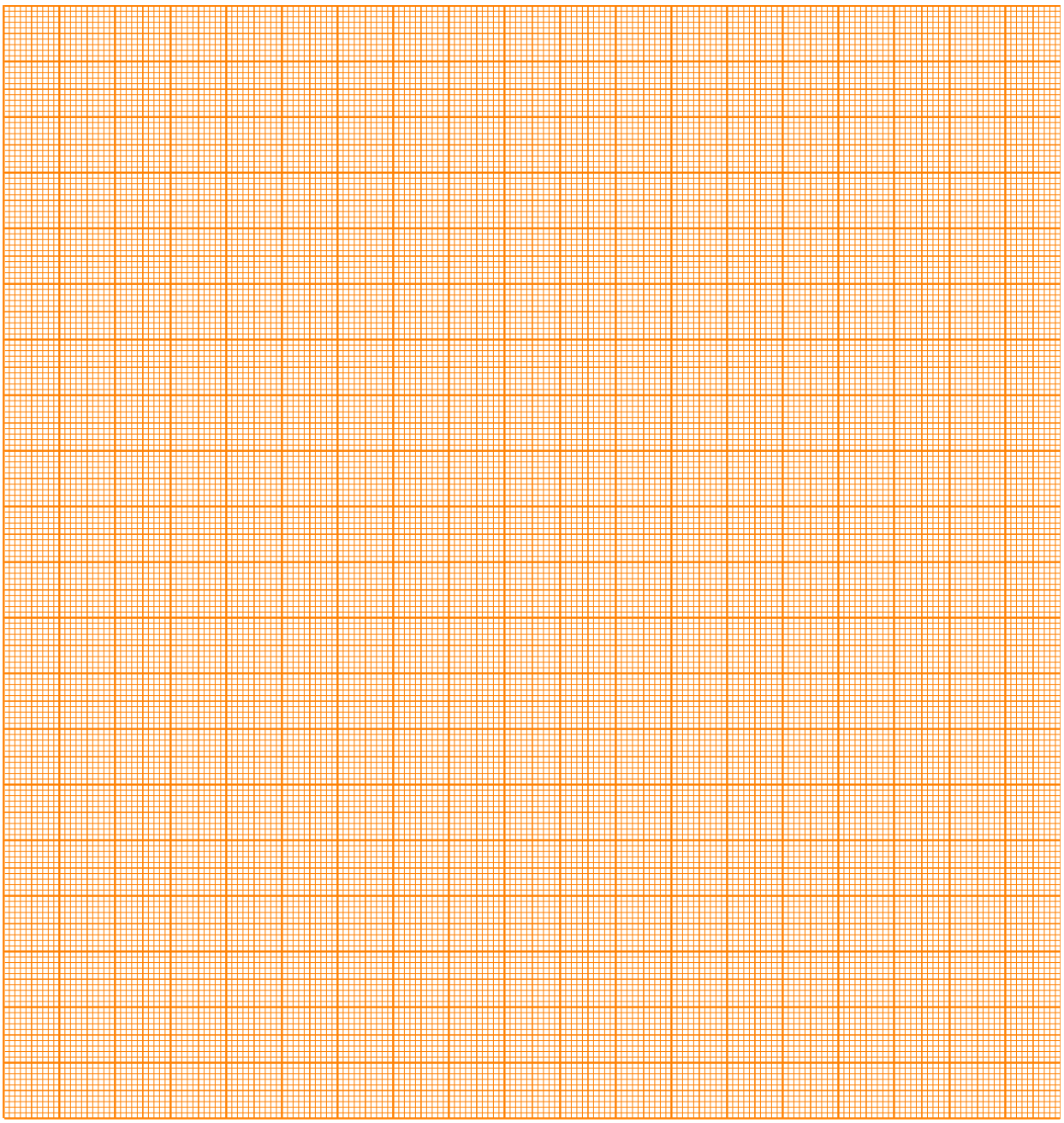
Measurements:

$\Delta z =$

**C.2** (3.5 pt)
Measurements:

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**C.2 (cont.)**
Graph:



**C.2 (cont.)**
Relationship plotted on graph:

Space to work

$n =$

## Part D: Spiked surface characterisation: spike formation and disappearance (3.1 points)

### Spiked surface characterisation: spike formation and disappearance

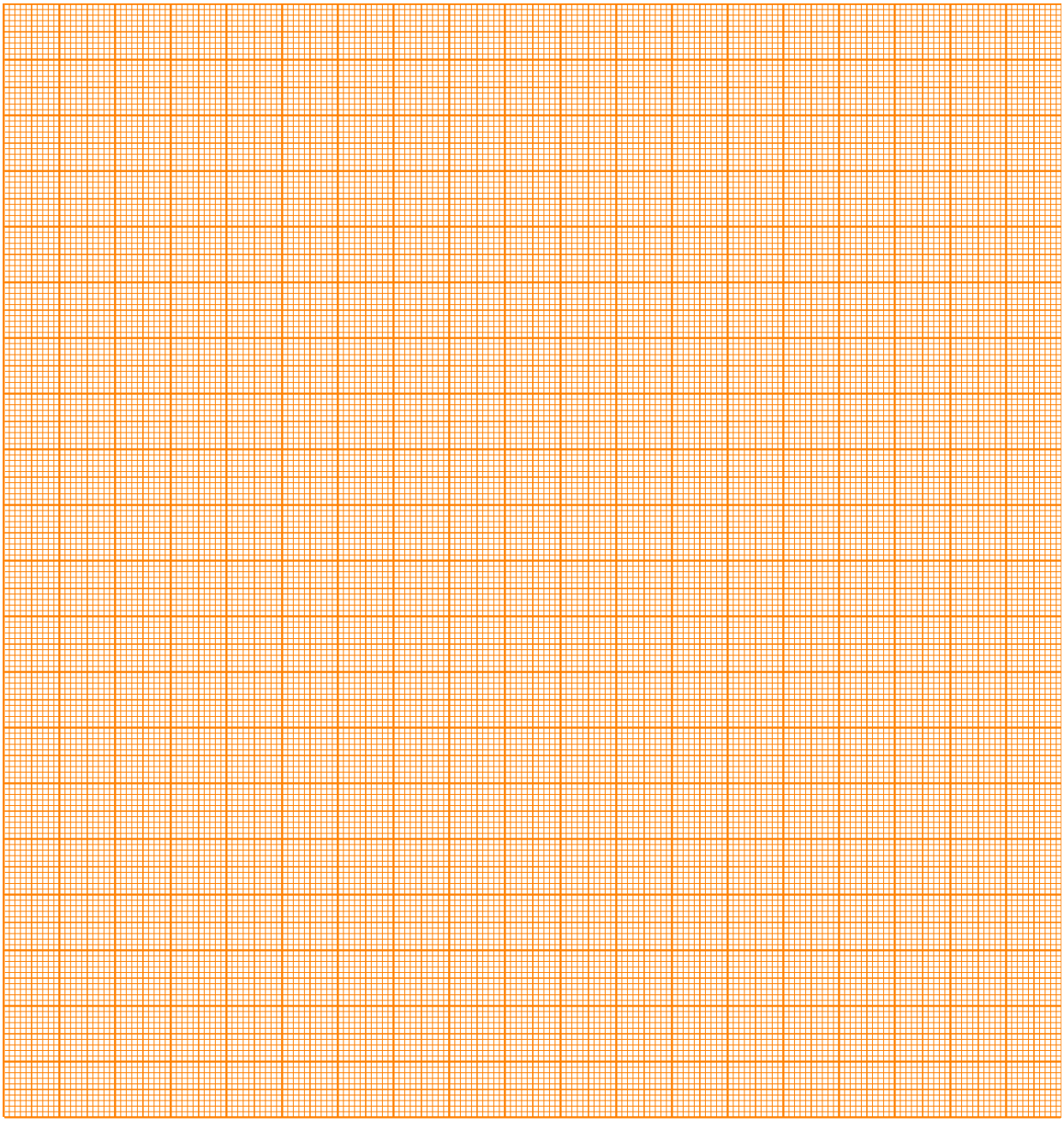
**D.1** (0.5 pt)

Surface tension of ferrofluid in air: $\sigma_{\text{fa}} =$

**D.2** (1.0 pt)

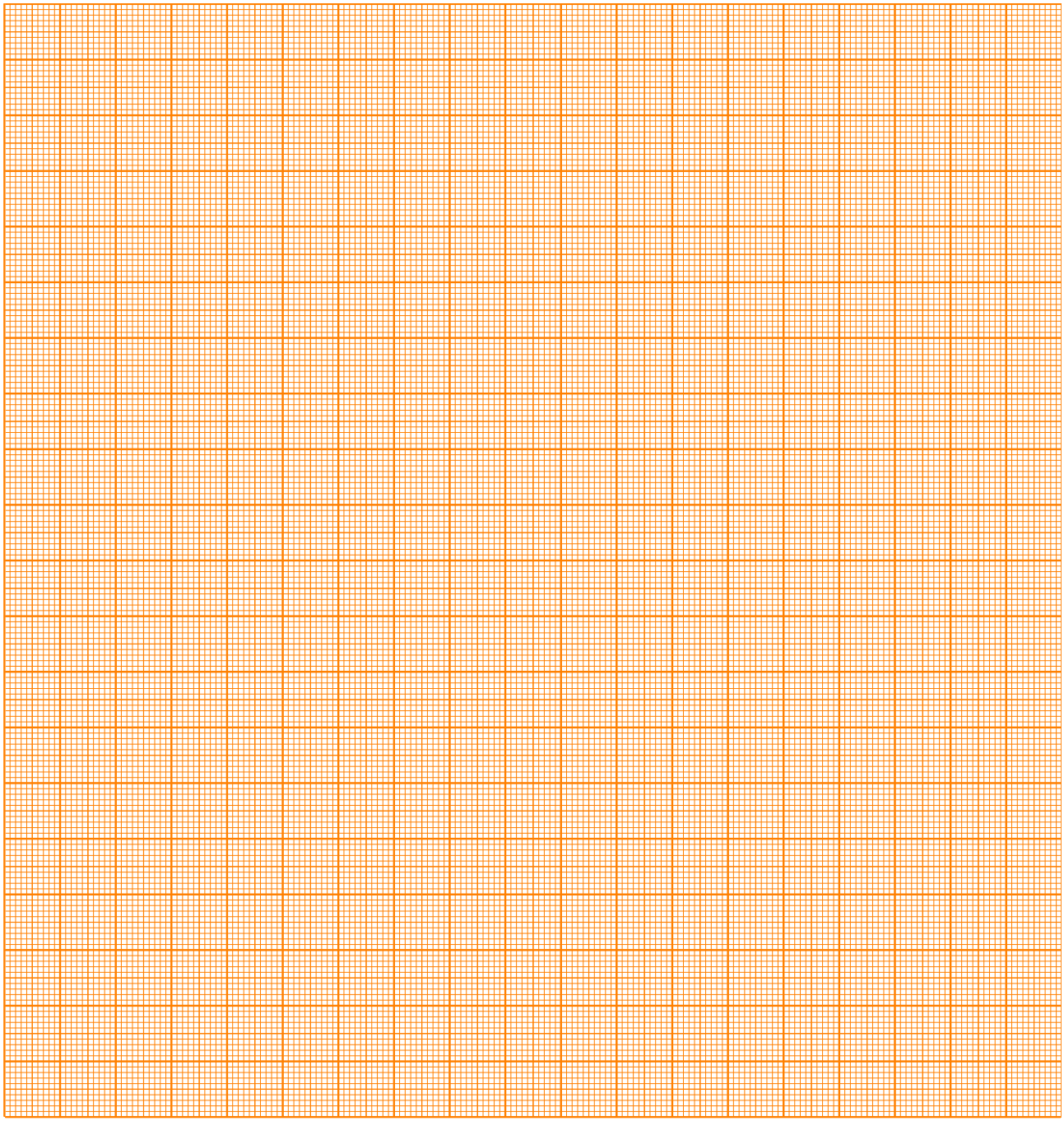
| Number of spikes | Magnet distance $z$ | $\Delta z$ |
|---|---|---|
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**D.3** (1 pt)



**D.4** (0.6 pt)
Sketch:

**Additional graph paper**



**Additional graph paper**

